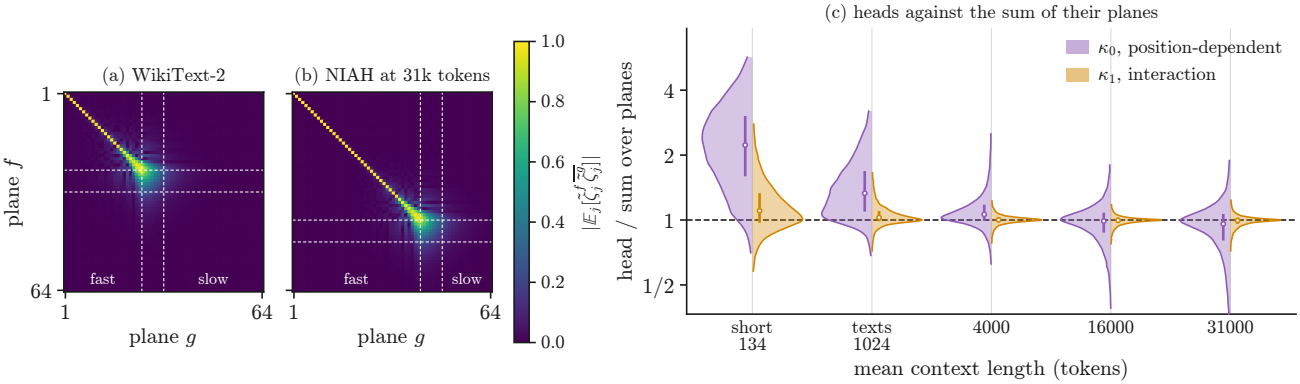

(a) WikiText-2
(b) NIAH at 31k tokens
plane f
plane g
1
64
fast
slow
$|\mathbb{E}_j[\zeta_j^f \overline{\zeta_j^g}]|$
0.0
0.2
0.4
0.6
0.8
1.0
(c) heads against the sum of their planes
κ_0, position-dependent
κ_1, interaction
head / sum over planes
1/2
1
2
4
short
134
texts
1024
4000
16000
31000
mean context length (tokens)